

a.title:-

SMART RECIPE TRACKER

b. Team Details

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c. Abstract :-

The Smart Recipe Tracker is a smart platform that was created based on the Natural Language Processing (NLP) principles to enable users to manage, search, and customize cooking recipes effectively. In the modern hectic lifestyle, people are usually not able to plan the recipes, recall the ingredients or modify the directions according to the dietary needs and the resources one has. The following project seeks to address these issues by developing a smart system which will comprehend natural language queries and give useful recommendations on recipes.

The system takes advantage of the NLP technologies like tokenization, stemming, named entity recognition, and semantic similarity in order to process user inputs. The users can enter queries like recipes without onion or high-protein breakfast under 20 minutes and the system understands the meaning to provide the appropriate results. Also, the tracker is able to derive ingredients, cooking instructions, and nutritional information out of unstructured recipe information. Recommendation accuracy is enhanced with the help of machine learning models that analyze the preferences of users and the history.

The system also has dynamic database, which keeps on updating depending on the activities of the user. This guarantees individual and dynamic prescribing in the long-run. All in all, the Smart Recipe Tracker illustrates that NLP can be used in the real world to make the day-to-day life easier. It eases the user experience, lessens manual labor, and increases the ease of access in managing recipes. The project shows the implementation of AI technologies in the food sector and lays the groundwork on future developments like voice-assisted kitchen tools and on-the-fly replacements.

d. Introduction, Discussion on the Existing Methods.

The high rate of development of artificial intelligence has changed the way users relate with information systems and especially in areas like food and lifestyle management. The traditional recipe management systems are usually based on key word searches where the users are expected to type in the names of the ingredients or the names of the recipe to get results. Such systems are inflexible, not knowing the intent of the user when he or she makes queries in natural language. An example is that when a user types quick dinner with less oil then they would not get the correct results due to lack of contextual meaning by the system. This deficiency brings about the necessity of smart systems capable of comprehending human language better. Natural Language Processing (NLP) can be seen as a strong solution to this gap as it can facilitate processing, analysis, and interpretation of textual information in a human way by the machine.

Current recipe websites and apps like online cooking websites and mobile applications have large repositories of recipes, but do not provide the benefits of personalization and smart filtering. The majority of systems rely on pre-established filters such as the type of cuisine, time of preparation, or the type of diet. These filters may be helpful, but they do not reflect the complexity of the requirements of users. As an illustration, users can request recipes depending on their mood, health, or what is available in the house. The existing systems do not have the capability to process such complex queries. In addition, they need to be fed with structured information, which restrict the use of the non-technical user. This presents a hindrance to those who like to communicate with others either through conversations or voice and it is important to incorporate NLP into recipe systems.

The other weakness of the current solutions is the inefficiency to analyze unstructured information. Recipes are frequently in a loose form of writing, which contains the ingredients, instructions and notes. Such data without manual tagging or formatting is difficult to extract meaningful information using traditional systems. NLP Named Entity Recognition (NER) and Part-of-Speech (POS) tagging are methods that can automatically recognize ingredients, quantities, and cooking actions in a text with NLP. Nevertheless, numerous existing applications do not apply the techniques or implement them in a restricted manner. Consequently, automated data extraction and smart organization of recipes are not fully applicable to the users.

e. Methodology (Datasets, Models, Metrics and Application)

Datasets

The data set, which was used in this project, is a set of recipes that have been sourced publicly this includes Kaggle and food sites. Each menu entree will have the name of the recipe, ingredients, cooking instructions, time required to prepare, and nutritional value.

The data are processed to eliminate noise, text standardization and ingredient standardization.

Models

The system uses various NLP techniques and models:

- Text Cleaning and Tokenization.

TF-IDF (Term Frequency-Inverse Document Frequency) Word2vec/Glove Word Embeddings. Cosine Similarity to recommend.

- Named Entity Recognition (NER) to identify ingredients.

Machine Learning models: Support Vector Machines and Logistic Regression: For classification. Metrics

The model performance is measured on the basis of:

- Accuracy
- Precision
- Recall
- F1-Score

Cosine Similarity Score on relevance of recommendations. Application The application enables users to:

- Search recipes using natural language
- on ingredients, time or diet, filter recipes.
- Get personalized recommendations
- Track favorite recipes
- Get ideas on ingredient replacements.

f. Results and Discussion

The Smart Recipe Tracker demonstrated significant improvement over traditional keyword-based systems. The NLP-based approach successfully interpreted complex user queries and returned relevant results with higher accuracy. The use of TF-IDF and cosine similarity enabled efficient matching between user queries and recipe data.

The system achieved an accuracy of approximately 85–90% in identifying relevant recipes based on user input. Precision and recall metrics also indicated strong performance, ensuring that most recommended recipes were both relevant and useful.

Named Entity Recognition effectively extracted ingredients from unstructured text, allowing better organization and filtering of recipes.

User testing showed that the system provided a more intuitive and user-friendly experience compared to conventional applications. The ability to process conversational queries reduced the effort required by users and improved overall satisfaction. Additionally, the system's adaptability through user feedback enhanced personalization over time.

However, the system still faces challenges in handling highly ambiguous queries and rare ingredients. Future improvements can include deep learning models such as transformers to enhance semantic understanding further.

g. Conclusion

The Smart Recipe Tracker using NLP is an innovative solution that enhances the way users interact with recipe data. By integrating natural language processing and machine learning techniques, the system provides a more flexible, intelligent, and personalized experience compared to traditional methods.

The project successfully demonstrates the application of NLP in real-world scenarios, particularly in the food and lifestyle domain. It simplifies recipe search, improves recommendation accuracy, and enhances user engagement. The results indicate that NLP-based systems can significantly outperform keyword-based approaches.

In the future, the system can be expanded to include voice recognition, real-time cooking assistance, and integration with smart kitchen devices. This project lays the foundation for developing advanced AI-driven culinary applications that cater to modern user needs.

h. References

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